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DSSAT modelling of conservation agriculture maize response to climate change in Malawi

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ABSTRACT

Adoption of conservation agriculture (CA) is increasingly being promoted as a way of adapting agricultural systems to increasing climate variability, especially for areas such as southern Africa where rainfall is projected to decrease. The DSSAT crop simulation models can be a valuable tool in evaluating the effects of CA which are viable both economically and environmentally. Our objectives were: (1) to evaluate the ability of DSSAT to predict continuous maize (*Zea mays* L.) yield for conventional tillage (CT) and CA systems as well as maize yield for a CA maize–cowpea (*Vigna unguiculata*) rotation on an Oxic rhodustalf (2) to use DSSAT to project weather effect of climate change on yield, economic returns and risk in CT and CA systems. The DSSAT model was calibrated using data from 2007–2008 season and validated against independent data sets of yield of 2008–2009 to 2011–2012 seasons. Simulations of maize yields were conducted on projected future weather data from 2010 to 2030 that was generated by RegCM4 using the A1B scenario. The DSSAT model calibration and validation showed that it could be used for decision-making to choose specific CA practices especially for no-till and crop residue retention. Long term simulations showed that maize–cowpea rotation gave 451 kg ha^{-1} and 1.62 kg mm^{-1} rain more maize grain yield and rain water productivity, respectively compared with CT. On the other hand, CT (3131–5023 kg ha^{-1}) showed larger variation in yield than both CA systems (3863 kg ha^{-1} and 4905 kg ha^{-1}). CT and CA systems gave 50% and 10% cumulative probability of obtaining yield below the minimum acceptable limit of 4000 kg ha^{-1} respectively suggesting that CA has lower probability of low yield than CT, thus could be preferred by risk-averse farmers in uncertain climatic conditions. Using similar reasoning, Mean-Gini Dominance analysis showed the dominancy of maize–cowpea rotation and indicated it as the most efficient management system. This study therefore suggests that CA, especially when all three principles are practiced by smallholders in the medium altitude of Lilongwe and similar areas, has the potential to adapt the maize based systems to climate change. Use of DSSAT simulation of the effects of CA was successful for no-till and crop residue retention, but poor for crop rotation. Refinement of crop rotation algorithm in DSSAT is recommended.

1. Introduction

Most climate change predictions using general circulation models (GCMs) generally show decreased precipitation trends for most of the southern Africa including Malawi. They also indicate more intense rainfall events and longer periods without rainfall, two factors generally attributed to global warming and thus greater risk for both floods and drought (IPCC, 2007). An analysis of climate risk by Lobell et al. (2008) indicated that identifying adaptation measures to reduce potential negative impacts of climate change on crops is important to a large food-insecure human population. Potential adaptation measures include improved germplasm with tolerance to drought and heat stress and improved soil and water conservation practices (Cairns et al., 2013). Conventional plant breeding has developed different crop species and/or varieties of the same crop that are more drought tolerant, but many of these materials have shorter growing seasons and thus offer a lower yield potential. Furthermore, according to Fischer and Edmeades (2010) genetic potential of increasing maize yields through breeding is limited because the room to further increase harvest index is small. Hence there is a need to develop integrated crop management solutions to adapt both varieties and cropping systems to climate change. To accomplish this across broad geographic areas, it is imperative to determine how different

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crop management systems will be affected by climate variability and how to select and implement the most viable systems on smallholders' farming systems.

Conservation agriculture (CA) is increasingly promoted as one way of adapting production systems to irregularities in rainfall patterns. CA is defined by three principles namely (i) continuous minimum mechanical soil disturbance, (ii) permanent organic soil cover and (iii) diversification of crop species grown in sequence and/or in associations (FAO, 2013). CA offers solutions to many of the problems associated with conventional tillage (CT) systems including low crop productivity, soil degradation and high production costs. Widespread adoption of CA by smallholder farmers in Malawi could help in overcoming the problem of soil degradation and adapting production systems to climate change thereby increasing rural household food security and reducing poverty (Malley et al., 2006).

Maize (*Zea mays* L.) is the major staple crop in southern Africa, including Malawi. In Malawi, it is mainly produced under rainfed conditions throughout the country on approximately 1.5 million hectares of land, accounting for 85% of total land under cultivation (MoAFS, 2012). Agriculture contributes 35% of the gross domestic product (GDP) (NSO, 2012). The demand of maize is likely to increase because of the increasing population pressure against limited production from available land as well as limited labour – emanating from migration of productive rural population to urban centres to seek for informal employment.

In Malawi, the dominant cultivation method is characterized by small handmade ridges consisting of ridge and furrow, a system that involves a lot of soil movement and high labour demand. This cultivation method is associated with clearing/burning of crop residues, monocropping and low use of organic fertilizers that contribute to increasing risks of soil erosion, depletion of soil organic matter, reduction of soil structural stability, soil fertility and soil water retention (Kumwenda et al., 1997; Zingore et al., 2005).

However, the effectiveness of CA to address the above mentioned problems largely depends on the capacity of change agents and farmers in applying the actual formulae and techniques of the principles of CA to local context (Wall, 2007). Short term to medium term potential benefits of CA have been assessed on farm in contrasting agro ecological zones of Malawi. CA generally results in increased yields especially after 3–5 years largely due to improved rainfall infiltration (Ngwira et al., 2012a,b; Thierfelder et al., 2013a); reduction in number of labour hours and increased returns to labour (Ngwira et al., 2012a,b); and reduction in risk of economic returns (Ngwira et al., 2013a,b). However, long term studies that report the long term effects of CA on crop productivity and net economic returns including risks are lacking in Malawi and southern Africa as a whole. At the same time, traditional experiments aimed at deriving appropriate cropping practices for the wide variety of soil types and climatic conditions are time consuming and expensive. One challenge is to develop methodologies that allow technology to be transferred effectively from one agro-ecological zone to another, given resource limitations that researchers face. In this context, use of crop simulation models (CSMs) is often considered useful to simulate different soil and crop management and climatic scenarios for developing the most suitable and site-specific strategies (Jones et al., 2003; Rezzoug et al., 2008). Computer-based assessment of different soil management practices and climatic scenarios ensures that only the most promising are selected for field testing in particular agro-ecological zones, thereby enhancing the efficiency of the research process. In Africa, modelling-assisted discussions among scientists and between scientists and farmers have provided a useful framework for designing field research for highly variable production environments; providing an opportunity for learning about new technologies and practices (Carberry et al., 2004) or exploring options for the sustainable intensification of production in smallholder farmers (Tittonell et al., 2009). Crop simulation models have become more useful with the incorporation of decision support systems that aid risk assessment and economic analyses of management strategies.

The Decision Support System for Agro-technology Transfer (DSSAT) (Jones et al., 2003) is a collection of several such models, which connects the decision support system to crop simulation models. DSSAT 4.5 has algorithms which can stimulate the influence of CA practices such as crop residue cover and tillage on soil surface properties and plant development. The other advantage of DSSAT 4.5 is that it has a separate program driver called Seasonal Analysis, which has the ability to analyse and compare the different management options biophysically and economically to guide choice of the most efficient management options (Thornton et al., 1998). DSSAT model has been extensively calibrated and validated across a number of environments in Malawi (Thornton et al., 1995) and will be used in this process on one particular component Crop Environment Resource Synthesis (CERES)-Maize. Scientists have used this model for making decisions in crop management under different environments and simulating the effects of climate change on crop production in Africa (Jones et al., 2003; Thornton et al., 1995, 2011). The model has also been used to assess the effects of CA on crop yields and soil water balance in China (Liu et al., 2013; Nangia et al., 2010).

The modelling reported in this study is one of the first studies applying the enhanced DSSAT model for evaluating the effects of CA systems in southern African region. In addition the paper addresses the question of whether modelling can add to the understanding of the advantages of CA over CT. Although CERES models of DSSAT has been used to predict yields of different crops in different soils and climatic conditions of several regions, simulation with CERES maize of DSSAT has not been tested for predicting the effects of CA on yield, risk and economic returns compared with CT. The aspect of crop simulation models for improving the efficiency of agricultural systems and to improve profitability including reducing risk was done using Seasonal Analysis program drivers in the CERES models of DSSAT. The present study was undertaken to identify management strategies for both ecologically and economically sustainable maize systems. This study tested the calibration and validation requirements of DSSAT using data sets of on station experiments.

Our objectives were: (1) to evaluate the ability of DSSAT to predict maize yield for CT continuous maize, CA (no-till and mulch) continuous maize and CA (no-till and mulch) maize–cowpea rotation on an Oxic rhodustalf (FAO, 1998) in the medium altitude agro-ecological zone of Malawi and (2) to assess the effect of climate change on yield, economic returns and risk in CT and CA using long term future projected weather data.

2. Materials and methods

2.1. Site description

Data for this study was collected from the long term CA trial located at Chitedze Research Station in Malawi. Chitedze is located on the mid-altitude agro-ecological zone of Malawi on the Lilongwe-Kasungu plain with a slope gradient of 2–6%, 1144 m above sea level at longitude 33° 39′ East and latitude 13° 58′ South. Chitedze has a mean temperature of 20 °C – maximum temperatures of more than 24 °C in November and lowest below 16 °C in July. The station receives a mean annual rainfall of about 900 mm, 85% which falls between November and March. Farmers in this region mainly grow maize as a monocrop, with those holding relatively large land areas practicing crop rotation i.e. rotating

Table 1
Treatments for long term conservation agriculture (CA) experiment at Chitedze Research Station, Lilongwe, Malawi, 2008–2013.

Check plot (CT), traditional farmers practice using the hand hoe (ridge and furrow system), maize as a sole crop, no residue retention, stubbles incorporated
CA basins (BA), maize as a sole crop, residue retention at a rate of 2.5–3 t/ha preferably maize stover
CA direct seeding (DS) with dibble stick, maize as a sole crop, residue retained as surface mulch
CA direct seeding (DS) with dibble stick, maize–cowpea rotation, residue retained as surface mulch, with both phases of rotation practiced every year
CA direct seeding (DS) with dibble stick, maize/pigeonpea intercropping, residue retained as surface mulch
CA direct seeding (DS) with dibble stick, maize/cowpea intercropping, residue retained as surface mulch
CA direct seeding (DS) with dibble stick, maize/mucuna intercropping, residue retained as surface mulch

maize with grain legumes and tobacco (*Nicotiana tabacum* L.). Chitedze soils have been classified as predominately Oxic rhodustalf (FAO, 1998).

2.2. Experimental design and crop management

The objective of this experiment, established in the 2007/2008 season, was to evaluate and monitor medium to longer term effects of CA practices on soil quality, weeds, pests and diseases and crop yield. The trial had seven treatments arranged in a randomized complete block design with four replicates per treatment (Table 1). Plot size consisted of 18 m $\times$ 13.5 m that was large enough to allow adequate within season sampling of various parameters. Tillage treatments studied were (1) CT of ridge and furrow system and (2) CA of no-tillage, direct seeding using dibble stick (Ngwira et al., 2012a,b). Previous years' crop residues were removed in the CT system during land preparation (emulating farmers' common practice) while these were retained in all CA plots. In CT plots, ridging was done using a hand hoe soon after harvest during the winter period. Weed control was done by hand hoes in CT while in CA, 2.5 L ha^{-1} glyphosate (N-(phosphono-methyl)glycine) was applied as a post-planting herbicide once weeds were present at seeding. Thereafter, upcoming new weeds were removed by hand weeding once they were 10 cm high or 10 cm in circumference to keep the fields weed free at all times. In all treatments, maize was seeded in rows 75 cm apart, one plant per station spaced at 25 cm apart within row aimed at a plant population of 53,333 plants ha^{-1}. All treatments received a uniform recommended fertilizer rate of 69 kg ha^{-1} N, 21 kg ha^{-1} P_2O_5 and 4 kg ha^{-1} S ha^{-1} applied as 23 kg ha^{-1} N, 21 ha^{-1} kg P_2O_5 and 4 ha^{-1} kg S ha^{-1} as basal fertilizer at planting and 46 kg ha^{-1} N in the form of urea 21 days after planting. Planting and harvest dates including anthesis dates and dates of physiological maturity during the six cropping seasons of the experiment are provided in Table 2. A medium maturing maize hybrid SC627 was used at the study site. This variety is tolerant to a range of maize diseases including maize streak virus (MSV), gray leaf spot (GLS) and *Turcicum* leaf blight; is moderately tolerant to low nitrogen conditions and is liked by farmers because it is white and semi-flint, which improves the poundability and storage qualities of the maize.

2.3. Harvest procedures

Maize was harvested after physiological maturity; yield was estimated from 10 sub samples of each 7.5 m^2 per treatment in the centre of the plot to avoid border effects. Grain was shelled by hand and separated from stover and cores. Grain was weighed using a digital scale and moisture content taken immediately to correct yields to 12.5% moisture. Cowpea (*Vigna unguiculata* (L.) Walp) in rotation with maize as well as intercropped cowpea, pigeonpea (*Cajanus cajan* (L.) Millsp) and velvet beans (*Mucuna pruriens* L.) were also harvested at physiological maturity. Yields of velvet beans and pigeonpea were very low due to competition for light, moisture stress and pest attacks, respectively. All maize stalks and leaves without cobs were weighed at harvest; 10 subsamples per plot were air dried at least 4 weeks before final dry weights were taken. Similarly all legume biomass were measured and recorded on dry mass basis. In this paper we report on maize yields only.

2.4. DSSAT model

DSSAT 4.5 was utilized for the study. DSSAT simulated crop growth, development and yield using a defined data set on crop management, minimum weather data and soil profile parameters. Some of the crop management data required to simulate DSSAT included crop, cultivar, planting date, row and plant spacing, fertilizer levels, tillage practices and organic amendments (Jones et al., 2003). Also included were data on physiological stages of crop growth such as anthesis dates, days to maturity and grain yield. Minimum weather data set consisted of maximum and minimum temperature, solar radiation and rainfall. Soil profile parameters included depth of soil, soil texture and chemical soil characteristics. In order to simulate yields under future climate scenarios, first the CERES-Maize model in DSSAT was calibrated and validated for the study. Climatic data were collected from the weather station located at Chitedze Research Station, less than 2 km away from the experimental site. Initial soil archives for the experimental site were developed and the weather variables were archived for every crop year. The monthly total values of rainfall used for the calibration and validation processes are presented in Table 3. The soil of the experiment was specified as sandy clay loam Oxic rhodustalf (FAO, 1998) in the model. The data on soil characteristics such as soil moisture, pH, bulk density, soil organic carbon and total-nitrogen were measured and stored in the soil input file (Table 4). The crop management data were recorded throughout the growing seasons. The input files, such as weather file, soil file, and A file (average measured data file), were prepared to calibrate and validate the CERES-Maize model.

2.5. Model calibration and validation

The algorithms in DSSAT 4.5 do not support CA basins and intercropping systems, therefore only three (CT continuous maize, CA continuous maize and CA maize–cowpea rotation) out of seven treatments were used to run the model. The process of calibration of the CERES-Maize aims at obtaining reasonable estimates of model genetic coefficients by comparing simulated data with the observed data. DSSAT model was run in its 'seasonal analysis' mode for this study. The model was calibrated using field measured

Table 2
Dates of planting, anthesis, physiological maturity and harvesting of the conservation agriculture experiment, Chitedze Research Station, Lilongwe, Malawi.

Activity	2007–2008	2008–2009	2009–2010	2010–2011	2011–2012	2012–2013
Planting	22-12-2007	25-11-2008	21-12-2009	09-12-2010	31-12-2011	11-12-2012
Anthesis	02-03-2008	29-01-2009	03-03-2010	17-02-2011	12-03-2012	19-02-2013
Maturity	13-05-2008	11-04-2009	14-05-2010	30-04-2011	23-05-2012	02-05-2013
Harvesting	22-05-2008	30-04-2009	07-06-2010	31-05-2011	03-06-2012	27-05-2013

Table 3
Rainfall (mm) data during calibration and validation period.

Month	2007–2008	2008–2009	2009–2010	2010–2011	2011–2012	Average
October	1.7	16.1	1.4	0.0	0.0	3.84
November	27.7	69.2	77.4	56.6	54.4	57.06
December	271.8	76.0	171.6	205.4	135.0	171.96
January	360.5	268.5	107.3	202.9	165.8	221
February	182.4	204.7	322.5	147.2	100	191.36
March	127.1	154.6	201.3	100.3	54.4	127.54
April	7.5	80.9	31.8	31.7	41.0	38.58
May	0.0	0.0	0.0	2.6	0.0	0.52
June	0.0	0.0	0.0	0.0	0.0	0.0
July	0.0	0.0	0.0	0.0	0.0	0.0
August	0.0	0.0	0.0	0.0	0.0	0.0
September	0.0	0.0	0.0	0.0	0.0	0.0
Total	978.7	870.0	913.3	746.7	550.6	811.86

values of weather parameters, crop management and soil properties during the 2007–2008 cropping season of the experiment. Genetic coefficients were estimated by using observed silking and maturity dates and grain yield of the maize variety SC627 for all treatments during the growing season with remaining seasons 2008–2009 to 2011–2012 used for subsequent model evaluation (Sarkar and Kar, 2006; Saseendran et al., 2013). An iterative approach was used to obtain reasonable genetic coefficients through trial and error adjustments until there was a match between the observed and simulated dates of silking and maturity and grain yield (Ma et al., 2006; Mavromatis et al., 2001). Values for maize cultivar MH16 archived in DSSAT 4.5 were used as baseline values to calibrate maize hybrid variety SC-627. MH-16 belongs to same maturity group as SC627 and was previously grown on the same research station and successfully calibrated and validated in the DSSAT model.

2.6. Seasonal analysis

Using the calibrated model, simulations were carried out with projected weather data of time trend of 20 years in the future from 2010 to 2030 to evaluate the effects of CA and CT practices on productivity as affected by climate change. Regional Climate Model (RegCM4) was used to project future climate data using the A1B emission scenario (Diro et al., 2012; Giorgi et al., 2012). The monthly total values of rainfall of the projected weather data used for the simulations are presented in Table 5. Future climate scenario indicated less rainfall during critical stages of maize growth i.e. flowering and grain filling than historical weather data of the site. While RegCM4 used in this study projected more rainfall at planting than the historical weather data, there was a reduction in the length of the season in the future. The model projected an average increase of about 2 °C of both maximum and minimum temperatures.

Seasonal analysis was used to compare CA and CT under soils of Chitedze and projected future weather scenarios. The program driver was run from 2010 to 2030 to select the best management options for maize crops. The three different treatment combinations were designated as three treatments of maize crops. Using long term rainfall data that varied in terms of dry, normal or wet years, the likely agronomic and economic effects including risk of CA and CT systems were determined and appropriate management decisions given. Variability in future weather data was assumed to describe uncertainty in seasonal weather. Thus, a distribution of yields and economic returns was produced, converting uncertainty in weather into uncertainty in yield and economic returns for different management options. Net return was estimated for each maize yield observation produced by each management system based on the 2010–2011 domestic maize price series and variable costs for each treatment (CT continuous maize, CA continuous maize and CA maize–cowpea rotation). Biophysical variable analysis was done using cumulative probability distribution (CPD) while decision criteria for economic factors and strategy analysis were done using Stochastic Dominance (SD) analysis and Mean-Gini Dominance (MGD) analysis, respectively. CPD is used to assess risk of management options and devise ways of reducing risk since future weather and prices are unknown. In CPD, the basic idea is to look at mean yield and variance at 0.5 cumulative probabilities. For a given defined minimum yield level, risk of each treatment was defined by looking at the cumulative probability of obtaining such a yield level. The treatment with highest cumulative probability was considered more risky. For Stochastic Dominance (SD) analysis, cumulative distribution function (CDF) was used to assess the riskiness of the treatments. In CDF analysis for two risky possibilities, A and B, A was accepted as dominant over B by first order stochastic dominance (FSD) if CDF of gains from A lied to the right of the CFD of B over the entire probability interval of 0 to 1. If the CDFs of A and B intersected, then no dominance by FSD could be established. However, if the area between the two CDFs below the point of intersection was greater than the area between the two CDFs above the point of intersection, then A dominated B by Second-Order Stochastic Dominance (SSD). SSD assumed that a decision maker is risk averse. Otherwise, no dominance was established, and both A and B were considered as second-order efficient. The results of strategic analysis were also confirmed by economic evaluation especially through Mean-Gini Dominance (MGD) analysis. MGD is an evaluation procedure in the seasonal analysis program, which calculated monetary return under each treatment combination and selected the most dominant treatment based on the highest economic return. In MGD, A dominated B if economic return of A (E(A)) was greater or equal to economic return of B (E(B)) and E(A) minus Gini Coefficient of distributions of A was greater or equal to E(B) minus Gini Coefficient of distributions of B.

2.7. Statistics

A generalized linear mixed model (GLMM) in GenSat 14.1 (VSN, 2011) was used to test effects of treatment, season and their interaction on maize yield during the six years (2008–2013) of the experiment. In the analysis, treatment was considered a fixed factor while season was considered a random factor. Treatment was considered fixed factor because it was specifically determined and its effect on maize yield was of major interest. Season was considered a random factor because it is assumed that season is randomly selected from a population of seasons with a normal distribution and a certain variance and was not of specific interest.

Table 4
Soil properties and initial conditions used for the DSSAT simulations.

Soil depth (cm)	%Sand	%Clay	Bulk density (g/cm^{-3})	pH (H_2O)	Total N (%)	OC (%)	Ca (cmol)	K (cmol)	Mg (cmol)	P (mg/kg)	Volumetric water (cm^3/cm^{-3})	NH_4^+ (g N/Mg soil)	NO_{3-}^{+} (g N/Mg soil)
10	55.66	37.41	1.39	5.07	0.23	2.70	7.84	0.32	1.92	15.06	0.20	0.431	2.84
20	54.56	39.56	1.40	5.03	0.17	1.92	7.56	0.32	1.80	12.01	0.21	0.272	1.14
30	53.16	39.28	1.39	5.07	0.13	1.51	7.31	0.26	1.81	9.59	0.22	0.171	2.16
60	53.34	38.53	1.34	4.96	0.05	0.65	7.12	0.27	1.87	5.84	0.20	0.203	2.61
90	54.78	37.66	1.36	4.97	0.02	0.24	6.74	0.2	1.91	4.02	0.21	0.132	1.48

Table 5
Comparison of historical (1980–2009) and projected (2010–2030) weather data at Chitedze Research Station.

	Historical weather data				Future weather data			
Month	SRAD	T_{max}	T_{min}	Rainfall (mm)	SRAD	T_{max}	T_{min}	Rainfall (mm)
January	18.3	27.0	18.2	221.2	20.3	29.3	19.7	134.2
February	18.3	26.9	17.9	196.6	19.9	29.5	19.7	124.5
March	17.9	27.2	17.3	188.9	18.8	28.4	19.2	152.5
April	17.0	26.8	16.1	70.1	21.2	28.8	17.1	48.9
May	16.1	26.4	13.2	14.7	23.7	28.7	14.8	5.7
June	15.5	24.9	10.4	1.1	24.3	26.7	11.9	0.3
July	15.4	23.8	9.2	0.4	25.1	26.3	10.9	0.6
August	17.8	24.8	9.8	0.6	27.9	28.3	12.0	0.2
September	20.3	27.5	11.7	0.8	29.8	31.2	15.1	0.8
October	22.2	29.5	14.6	6.1	30.0	33.2	18.0	14.4
November	22.6	30.6	16.9	38.7	23.6	31.3	20.1	83.2
December	20.1	29.0	18.1	140.1	20.6	29.0	19.5	188.6
Average	18.5	27.0	14.5	879.3[a]	23.8	29.2	16.5	753.9[a]

SRAD means solar radiation (MJ/m^2 d), T_{max} is maximum temperature (°C), T_{min} is minimum temperature (°C).
[a] Average rainfall annual totals.

However, the major interest of the seasonal effect was on the variation among them.

The statistics used for the performance evaluation of DSSAT model were Root Mean Square Error (RMSE) and mean percent difference (MPD) which was obtained as the mean of the %*D* (the difference between the predicted (P_i) and observed (O_i) (Ahuja et al., 2000). The value of RMSE equal to zero indicated the goodness of fit between predicted and observed data.

$$\text{RMSE} = \left[\frac{\sum_{i=1}^{n}(P_i - O_i)^2)}{n}\right]^{0.5} \quad \text{and} \quad \%D = \left[\frac{O_i - P_i}{O_i}\right] \times 100$$

3. Results and discussion

3.1. Maize grain yield during the experimentation period

Maize grain yield averaged across season showed significant ($p < 0.05$) differences between treatments. CA maize–cowpea rotation gave 1335 kg ha^{-1} greater yield than CT continuous maize. Maize grain yield showed no significant differences between treatments in the first four seasons (Table 6). In the fifth season, CA maize–cowpea rotation gave 1953 kg ha^{-1} greater maize grain yield than CT continuous maize. In the six season, CA maize–cowpea rotation, CA continuous maize, CA maize–cowpea intercropping and CA maize–velvet bean intercropping gave 3012, 1370, 1819 and 1481 kg ha^{-1}, respectively more maize yield than CT continuous maize. However, CA basins and CA pigeonpea intercropping gave similar yields to CT continuous maize. Highest yield observed in CA maize–cowpea rotation plots could be attributed to a combined effect of multiple factors including reduced pest and weed infestations, improved water use efficiency, good soil quality (higher SOC) and greater biological activity (Nyamangara et al., 2013; Thierfelder et al., 2012, 2013b). The results confirm the time lag before farmers could expect significant benefits by adopting CA in this Lilongwe plain. However, unlike earlier on-farm studies (Ngwira et al., 2012a,b; Thierfelder et al., 2013a) that reported increased yield both in CA continuous maize and CA maize–legume intercropping systems from fourth season,

Table 6
Maize grain yield in different conservation agriculture systems and one conventional ridge and furrow system, Chitedze Research Station, Malawi, 2008–2013.

System	Crop	2007–2008	2008–2009	2009–2010	2010–2011	2011–2012	2012–2013	Mean
CT	Maize	4832a	4568a	4964a	6321a	5118b	5057c	5143b
CA Basins	Maize	4700a	4404a	4725a	6773a	4228b	5626bc	5076b
CA	Maize	5259a	4589a	4780a	6365a	4906b	6427b	5388b
CA rotation	Maize–cowpea	5545a	4272a	5766a	8144a	7071a	8069a	6478a
CA intercrop	Maize + pigeonpea	5322a	4506a	5123a	7095a	5101b	6341bc	5582b
CA intercrop	Maize + cowpea	4577a	4318a	4533a	6081a	4414b	6876ab	5133b
CA intercrop	Maize + mucuna	4586a	3853a	4819a	6763a	4825b	6538b	5231b
	LSD ($p \leq 0.05$)	1082	1483	928	1507	1611	1293	
	F-statistic	1.06	0.23	1.5	1.66	2.96	4.78	
	p-Value	0.415	0.961	0.228	0.18	0.03	0.004	

Values followed by the same letter within each column in the same year are not significantly different from each other.

Table 7
Calibrated genetic plant growth coefficients of maize variety SC627 used in CERES-Maize model at Chitedze Research Station, Lilongwe, Malawi.

Parameter	Value
P1: Thermal time from seedling emergence to the end of juvenile phase (expressed in degree days above a base temperature of 8 °C) during which the plant is not responsive to changes in photoperiod.	230.0
P2: Extent to which development (expressed as days) is delayed for each hour increase in photoperiod above the longest photoperiod at which development proceeds at a maximum rate (which is considered to be 12.5 h)	0.6
P5: The thermal time from silking to physiologixal maturity (expressed in degree days above a base temperature of 8 °C)	940.0
G2: Maximum possible number of kernels per plant	430.0
G3: Kernel filling rate during the linear grain filling stage and under optimum conditions (mg/day)	6.0
PHINT: Phylochron interval; the interval in thermal time (degree days) between successive leaf tip appearances	38.9

Table 9
Validation data.

Year	Treatment	Observed	Predicted	Error (%)[a]
2008/09	CT sole maize	4568	3923	−14
2008/09	CA DS sole maize	4589	4360	−5
2008/09	CA DS maize–cowpea rotation	4272	4358	2
RMSE (kg ha^{-1})		295		
MPD		3.9		
2009/10	CT sole maize	4964	4920	−1
2009/10	CA DS sole maize	4780	5009	5
2009/10	CA DS maize–cowpea rotation	5766	5032	−13
RMSE (kg ha^{-1})		416		
MPD		1.7		
2010/11	CT sole maize	6321	3955	−37
2010/11	CA DS sole maize	6365	4947	−22
2010/11	CA DS maize–cowpea rotation	8114	4946	−39
RMSE (kg ha^{-1})		2129		
MPD		30		
2011/12	CT sole maize	5118	4661	−9
2011/12	CA DS sole maize	4906	4882	0
2011/12	CA DS maize–cowpea rotation	7071	4881	−31
RMSE (kg ha^{-1})		974		
MPD		5.8		

[a] Negative values for error (%) indicate that the model underestimated maize grain yield.

the on-station studies did not find any significant differences on maize yield between such treatments and CT before the first five seasons. In general, farmers' fields are characterized by poor soils largely due to long-term historical monocropping and insufficient return of organic material to the soil, which could lead to greater improvements in soil quality due to adoption of CA. Hence their effects on crop yield are more evident in the medium term. On the other hand, Chitedze Research Station, has favourable soil and rainfall conditions for plant growth (one of the principle reasons to place a maize breeding station), which suggests it would take more time to observe significant yield differences between CT and systems with partially applied CA principles. Similar results have been reported in a study comparing soil in contrasting environments of Zimbabwe where soil quality improvements in the early stages of CA implementation were more evident on degraded soils compared with soil with greater fertility (Thierfelder and Wall, 2012). This suggests that in favourable environments, significant differences between CT and CA systems can only be realized if all principles are applied, which will improve soil quality in the medium term than partial application of CA principles. Similar results have been reported in a meta-analysis of CA systems done by (Rusinamhodzi et al., 2011) where the benefits of CA on maize yield were only apparent after several years.

3.2. DSSAT calibration and validation

The variables used for calibration were anthesis date, maturity date and maize grain yield (2007–2008). The seven genetic coefficients of cultivar SC627 of maize, which were calculated in the present study, are presented in Table 7. The range of these parameters lies within the values reported by other researchers for medium maturing maize varieties (Liu et al., 2013; Nangia et al., 2010; Thornton et al., 1995). The calibration process revealed that the model predicted maize grain yield 'well' as the mean difference between simulated and observed values was found to be 2.6% and RMSE was 400 kg ha^{-1} (Table 8). This implies that the model was successfully calibrated for the three treatments in question. There was generally 'good' agreement between predicted and observed anthesis date and date of physiological maturity as the error was very low for all the treatments. The error in predicting yield for all treatments was below 12% which was considered 'good' (Bakhsh et al., 2013; Liu et al., 2013; Nangia et al., 2010).

The CERES-Maize model was evaluated by comparing simulated and observed yield data for 2008–2009, 2009–2010, 2010–2011 and 2011–2012 growing seasons for all the three treatments used in DSSAT (Table 9). There was a good agreement between observed and simulated grain yield data, especially for 2008–2009; 2009–2010 and 2011–2012 seasons. The model simulated maize grain yield well for all treatments with difference ranging from −14 to 2% for 2008–2009 growing season, −13 to 5% for 2009–2010 growing season and −31 to −9% for 2011–2012 growing season (except for CA maize–cowpea rotation during this season). There were high errors (−39 to −22%) in predicting yield during 2010–2011 season which we cannot explain. Overall, the RMSE was found to be as 295 kg ha^{-1}, 416 kg ha^{-1}, 2129 kg ha^{-1} and 974 kg ha^{-1} for 2008–2009, 2009–2010, 2010–2011 and 2011–2012 seasons, respectively. Similarly, MPD was found to be as 3.9%, 1.7%, 30% and 5.8% for 2008–2009, 2009–2010, 2010–2011, and

Table 8
Calibration data.

Treatment	Anthesis date			Physiological maturity			Maize grain yield (kg ha^{-1})		
	Observed	Predicted	Error (%)	Observed	Predicted	Error (%)	Observed	Predicted	Error (%)
CP sole maize	70	71	1	140	141	1	4832	4271	−12
CA DS sole maize	70	71	1	140	141	1	5259	5002	−5
CA DS maize-cowpea rotation	70	71	1	140	141	1	5545	5190	−6
RMSE (kg ha^{-1})							400		
MPD							2.6		

Negative sign represents under prediction.

Table 10
Predicted yield and rainfall productivity of maize as influenced by tillage and crop residue retention, 2010–2030.

Treatment	Maize grain yield ($kg\,ha^{-1}$)		Rainfall productivity (kg yield/ha/mm rain)	
	Mean	SD	Mean	SD
CT-sole maize	3937.2b	465.6	7.94b	0.58
CA-sole maize	4335.8a	227.6	9.64a	0.42
CA maize–cowpea rotation	4388.3a	242.7	9.56a	0.41
p-Value	<0.001		<0.001	

Values within the same column followed by the same letter are not significantly different from each other.
SD means standard deviation.

2011–2012 seasons, respectively. This comparison shows that the model has the potential to simulate maize yield for an independent data set of the year. However, the higher negative errors for CA maize–cowpea rotation especially during 2010–2011 and 2011–2012 seasons suggest that the yield of maize was underestimated with increasing duration of practicing CA increased. High prediction errors were also found in no-till and crop rotation systems under 'black soil' in China (Liu et al., 2013). This is possibly due to lack of accounting in DSSAT for the cumulative soil benefits of crop rotation systems thereby under predicting maize grain yield. DSSAT uses same initial soil conditions for the entire simulation period; hence refinements of the soil changes algorithms in DSSAT is recommended in order to sufficiently and accurately predict yield of crop rotation systems.

Overall, the results showed that the performance of CERES-Maize was acceptable under a given set of conditions. Similar results have been reported by Liu et al. (2013), Nangia et al. (2010) and Sommer et al. (2007), therefore the model can be used further for decision-making regarding CA practices especially for no-till and crop residues retention but has its limitations when predicting the likely outcome of maize–legume rotations.

3.3. Predicted grain yield and rainfall productivity

The mean grain yield of maize, as predicted by seasonal analysis (2010–2030), was significantly greater in CA maize–cowpea rotation and CA continuous maize than CT continuous maize (Table 10). The increase of yield and rain water productivity followed the same pattern in CA with no-till and crop residue retention. The mean yield and rain water productivity was highest under CA maize–cowpea rotation, which gave 0.45 $Mg\,ha^{-1}$ more yield and 1.62 $kg\,mm^{-1}$ rain more rainfall productivity than CT. Prediction errors for maize yields and rainfall productivity were low.

Simulations conducted from 2010 to 2030 using projected weather data predicted that maize yields varied with seasonal rainfall with greater variation observed in CT fields. While the largest variation in yield in CA treatment was from 3863 $kg\,ha^{-1}$ to 4905 $kg\,ha^{-1}$, yield of CT varied from 3131 $kg\,ha^{-1}$ to 5023 $kg\,ha^{-1}$ (Fig. 1). These results suggest that CA practices have the ability to stabilize maize yields in uncertain rainfall patterns. Both CA treatments always had similar and greater yields than CT continuous maize. The differences in crop yields between CA treatments and CT were generally small in high rainfall seasons but were much larger in low rainfall seasons suggesting yield advantage of CA under low rainfall compared with CT (Fig. 1). The projected rainfall by RegCM4 model indicated lower amounts of rainfall during critical periods of maize growth compared with historical long-term rainfall average which suggest that CA had the ability to enhance maize crop capacity to withstand negative effects of dry spells. This shows that despite DSSAT failure to take into account the accumulated soil benefits of CA maize–cowpea rotation in long term simulations, the model was very sensitive to predict changes in yields due to no-till and mulching benefits of CA compared with CT, especially during dry years. This implies DSSAT was able to take into account the effects of a combination of no-tillage and mulch cover on soil water balance. This model sensitivity is due to recent incorporation of tillage and organic amendments algorithms in DSSAT 4.5. Similar DSSAT simulation

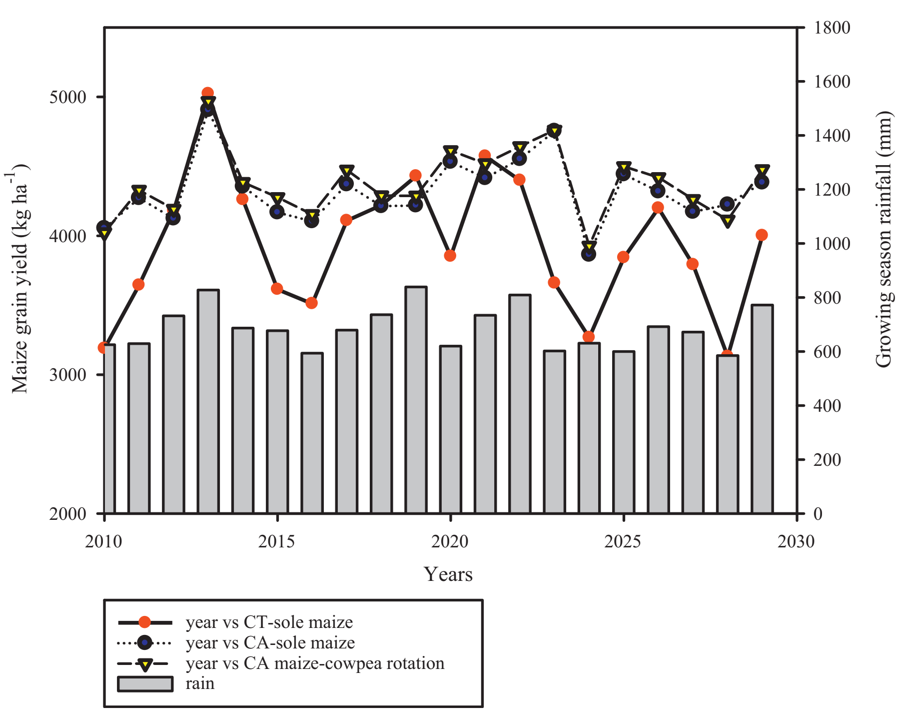

Fig. 1. Comparison of predicted crop yields for conventional tillage, CA directed seeded sole maize and CA directed seeded maize-cowpea rotation during 2010–2030.

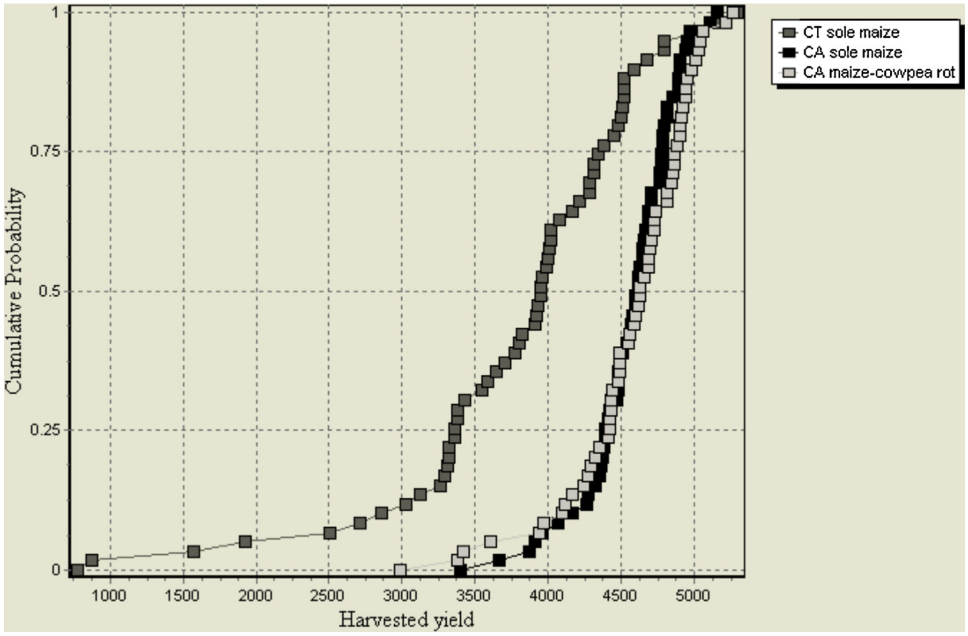

Fig. 2. Cumulative probability distribution plots on maize yields for different treatments.

results have been reported where no-till with whole maize stalk mulching resulted in 17–23% greater yield than CT in a dry spring in Shouyang County in China and the researchers attributed this yield advantage to early plant establishment (Cai and Wang, 2002).

3.4. Selection of management options through strategic analysis and economic analysis

The plots of cumulative probability distribution (CPD) at 0.5 showed that the highest mean yield was predicted under both CA systems and lowest under CT (Fig. 2). This pattern was consistent even for the predicted mean variance of yield which was smaller for CA continuous maize followed by CA maize-cowpea rotation and maximum for CT suggesting that CT could be regarded risky enterprise among risk averse farmers. Given a yield target of 4 Mg ha^{-1} calculated to satisfy a healthy diet (Peter and Herrera, 1989) of an average family of 5.4 in Malawi (Ngwira et al., 2013a,b), from the CPD it can been shown that CT and both CA systems gave 50% and 10%, respectively cumulative probability of obtaining yield below the minimum acceptable limit. This suggests that there is high probability of CT to give yield below the minimum acceptable yield target than CA systems.

Similarly stochastic dominance analysis showed that both CA systems were less risky than CT since these lied to the right of CT in the CPD plots, with both CA systems having lowest variance in monetary returns (Fig. 3). Mean-Gini Dominance (MDG) analysis

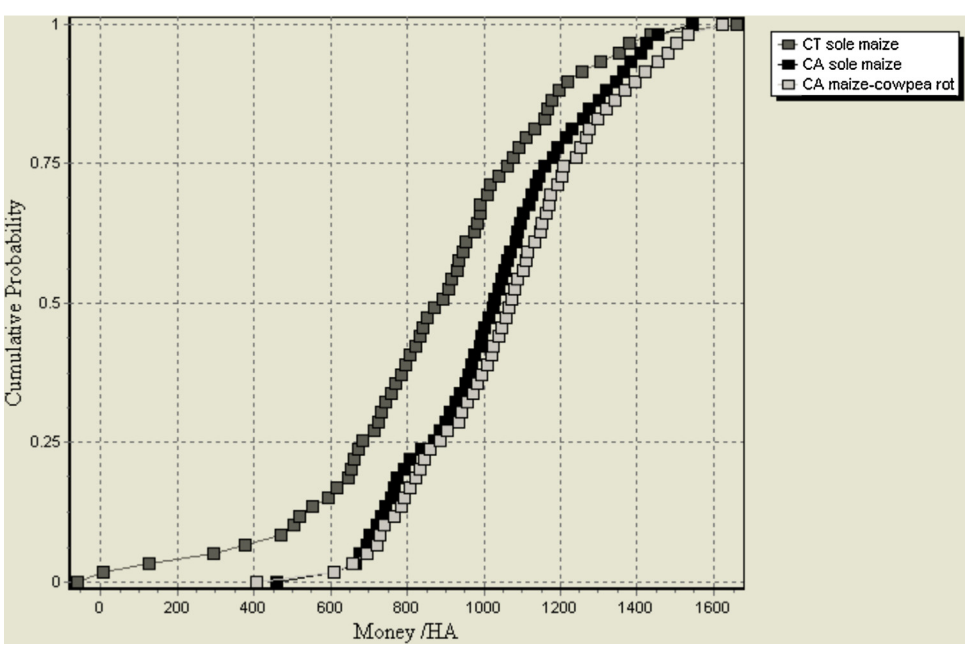

Fig. 3. Stochastic dominance analysis of monetary returns ($/ha) of maize under different treatments.

Table 11
Dominance analysis of different tillage management strategies of maize.

Treatment	$E(x)$	$E(x) - F(x)$	Efficient (yes/no)
CT sole maize	860.6	678.2	No
CA DS sole maize	1021.5	886.7	No
CA DS maize–cowpea rotation	1063.1	918.8	Yes

Mean-Gini Dominance: $E(x)$ mean return \$/ha.
Mean-Gini Dominance: $F(x)$ Gini coefficient \$/ha.

showed the dominancy of maize–cowpea rotation and indicated it as the most efficient management system (Table 11).

Previous on-farm studies in Malawi have shown the feasibility of adopting no-till and residue retention by farmers (Ngwira et al., 2012a,b; Thierfelder et al., 2013a). This means that for risk averse farmers, they are better off adopting CA than the status quo. However, given the biophysical benefits and potential of CA systems to reduce risk, will farmers in Malawi adopt crop rotation? In addition to being main staple, maize has become a political crop where policy interventions such as farm input subsidies are geared towards improving production of the crop among smallholder farmers who prioritize food security concerns above other needs amidst small land holdings (Thierfelder and Wall, 2010; Thierfelder et al., 2012; Umar, 2013). Therefore, integration of crop rotation in their production systems will require innovative ways that are compatible with their immediate farming objectives. Given that farmers in Malawi posses small land holdings, intensification of maize production in less area of land would help to release part of the land to the production of other crops in rotation with maize. However, the choice of the crops in rotation will depend on other factors such as access to seed and output market which are cited as major bottlenecks to crop diversification in smallholder farms in southern Africa (Umar et al., 2011). One proposed option to address market constraints is to link farmers to existing stockists, commodity chains such as Department for International Development (DFID) Research into use (RIU) legume (beans, soybeans, pigeonpea, etc.) platforms and farmer organizations such National Smallholder Farmers Association of Malawi (NASFAM). Provision of access to seed and produce market can act as an incentive for farmers to invest in more balanced cereal–legume rotation that will enable farmers realize economic benefits, reductions in the risks and uncertainty of production.

4. Conclusion

This research study showed positive benefits over 5 years of no-till systems if accompanied with sound crop rotation/intercropping and crop residue retention. The DSSAT model calibration and validation was successful especially for no-till and mulch but poor for crop rotations. This highlights the need to develop a rotation module into DSSAT to predict future effects of crop rotations. Long term future climate simulations showed that CA was less vulnerable to climate variability expressed by higher yields in drier years than CT. Similarly, cumulative probability distribution indicated that CT was more a risky enterprise compared with CA systems. Using similar reasoning Mean Gini-Coefficient identified CA maize-cowpea rotation as the most efficient management system, and therefore could be preferred by risk averse farmers.

This study has shown that adoption of CA in medium altitude areas of Lilongwe may prepare smallholder farmers for the coming future threats of climate variability and change. While application of the full principles of CA indicated benefits in terms of less vulnerability to lower yields in dry years, successful implementation of CA by smallholder farmers will depend on how other factors such as input and output market of grain legumes, linking farmers to farmer organizations among others are addressed.

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